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Strain-Data-Driven Force Reconstruction Using Pseudo-Inverse Matrix



Aditya Panigrahi, Alexander Q. Nguyen, Marc A. Eitner, and Jayant Sirohi

Abstract This chapter presents a comprehensive study that focuses on developing and rigorous assessment of a predictive model tailored for estimating bending loads within structural systems. The model leverages strain measurements as input parameters, processed through the pseudo-inverse matrix technique, resulting in highly accurate load predictions. The investigation explores the model's performance across various scenarios, with particular emphasis on its sensitivity to noise and its impact on prediction accuracy. This chapter highlights the model's sensitivity to point load placement (L_f), demonstrating that shifting the point load away from the boundary condition reduces load sensitivity, while positioning it closer heightens sensitivity, making it more susceptible to noise. Moreover, the research investigates scenarios involving sensor removal and malfunction, showing their substantial impact on load prediction. Additionally, the study extends the model's capabilities to predict temperature changes based on thermal strain inputs, achieving close alignment with actual values. In summary, this predictive model proves robust in estimating bending loads, but its sensitivity to noise, load placement, and the importance of sensor quality are emphasized, along with its versatile application in temperature change prediction.

Keywords Load estimation · Machine learning · Pseudo-inverse · Analytical model

Introduction

In aerospace technology, the integration of Machine Learning (ML) and Artificial Intelligence (AI) has ushered in an era marked by innovation and enhanced operational efficiency. Seminal contributions, such as Zhang et al. [1], have played a pivotal role in advancing the capabilities of unmanned autonomous systems, propelling trends in enhanced perception, precise control, and automated flight path optimization. Furthermore, ML applications in additive manufacturing for aerospace [2], optimizing topology, generative design, and product resolution ultimately reduce weight and elevate manufacturing quality. Lastly, in simulations and training, ML techniques, such as physics-informed learning and data-driven modal decomposition [3], play a pivotal role in optimizing fluid dynamics, mitigating jet noise, and simulating turbulent combustion under specific conditions. These references underscore the transformative impact of ML in the aerospace industry, promising innovation and safety advancements for aviation's future.

Another application of integrating predictive models into aerospace is the reconstruction of load based on a network of internal sensors. The precise reconstruction of forces acting on structures is an enduring challenge, necessitating the application of innovative regularization techniques, especially in scenarios involving ill-conditioned matrices and the presence of noise. Building upon the work of Ocejó and De Smet [4], who introduced multiplicative regularization as an alternative to computationally intensive Tikhonov-like additive regularization, X. Sun et al. [5] introduced novel regularization approaches for dynamic load identification. These methods leverage improved regularization operators, the L-curve method, and experimental validations on a vehicle hood to offer more accurate load approximations than traditional Tikhonov regularization. Furthermore, the work of W. Gao and K. Yu introduces a novel method based on quadratic programming theory for determining the Tikhonov regularization parameter [6], overcoming challenges associated with ill-conditioned matrices and promising enhanced accuracy, stability, and noise resistance in load identification.

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Additionally, R. Guo et al. explore combining Singular Value Decomposition (SVD) and Tikhonov Regularization to solve ill-conditioned load identification problems. Their study concludes that Tikhonov regularization is suitable for condition numbers exceeding 1000, while SVD excels in other cases, showcasing the potential for improved load identification in engineering applications. Furthermore, H.G. Choi, A.N. Thite, and D.J. Thompson introduce a threshold-based approach for selecting between pseudo-inversion and Tikhonov regularization [7]. Based on condition numbers and matrix dimensions, this threshold calculation offers guidance for optimizing regularization methods. In the pursuit of advancing iterative solutions, Linjun Wang et al. present an improved iterative regularization scheme for solving inverse problems [8]. This scheme outperforms conventional methods, providing engineers with a more robust toolset for transforming data from strain and Fast Fourier Transform (FFT) into force information.

Lastly, the analysis and regularization of a deconvolution problem are explored [9]. This problem involves determining the impact response of a system, with SVD playing a crucial role in analyzing singular values to optimize solutions. Noise mitigation and determining truncation points in SVD are vital components in tackling this challenge.

This chapter aims to develop and rigorously assess a predictive model tailored for estimating bending loads of a benchtop model of a hypersonic vehicle. The model's core relies on strain measurements as input parameters, processed through the pseudo-inverse matrix technique for load predictions. The research investigates the model's performance across various scenarios, particularly emphasizing its sensitivity to noise and its impact on results. An essential aspect of this work is creating an analytical beam model based on the Euler–Bernoulli beam theory. The study also examines the number of training datasets required for point load predictions and assesses the model's robustness in the presence of synthetic noise in the strain data. Additionally, it explores the importance of point load placement and data quality in load prediction. In an extension of the model's capabilities, the study explores its potential in predicting temperature change. In summary, this research establishes a robust predictive model for estimating bending loads and assesses its sensitivity to various factors.

Methodology

Initial Concept 3x (IC3x)

The Initial Concept 3x (IC3x) vehicle, initially created from a sizing study by the AFRL Munitions Directory, and then used by Dr. Ryan Klock in his dissertation [10], is a rocket-scamjet two-stage vehicle focusing on post-booster jettison operation. Its length is 3.56 m, transitioning from a pointed nose to a cylindrical fuselage for optimized atmospheric ascent. The IC3x features diamond airfoil fins with maximum thickness at 50% chord, augmented by a 1 mm Thermal Protection System (TPS) layer. The leading edges of the fins exhibit a 67.3-degree sweep angle, while the trailing edges remain unswept, ensuring a balanced combination of stability and maneuverability for high-speed, high-altitude applications in aerospace research and exploration.

For the purposes of experimentation, the vehicle was scaled down to the length of 40% of the full-scale model, a version that could be used to conduct static load experiment. Figure 1 displays this experimental setup. The experimental model, the

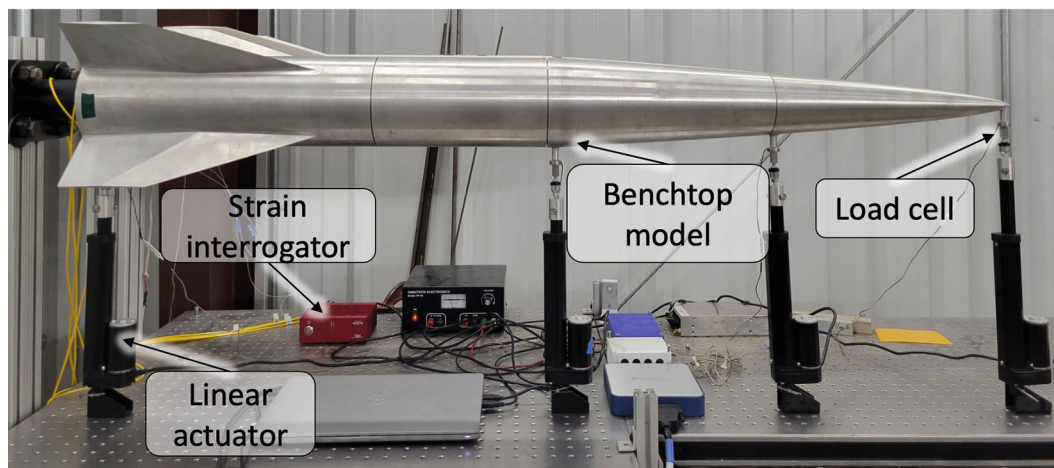


Fig. 1 Experimental setup of IC3x-BM

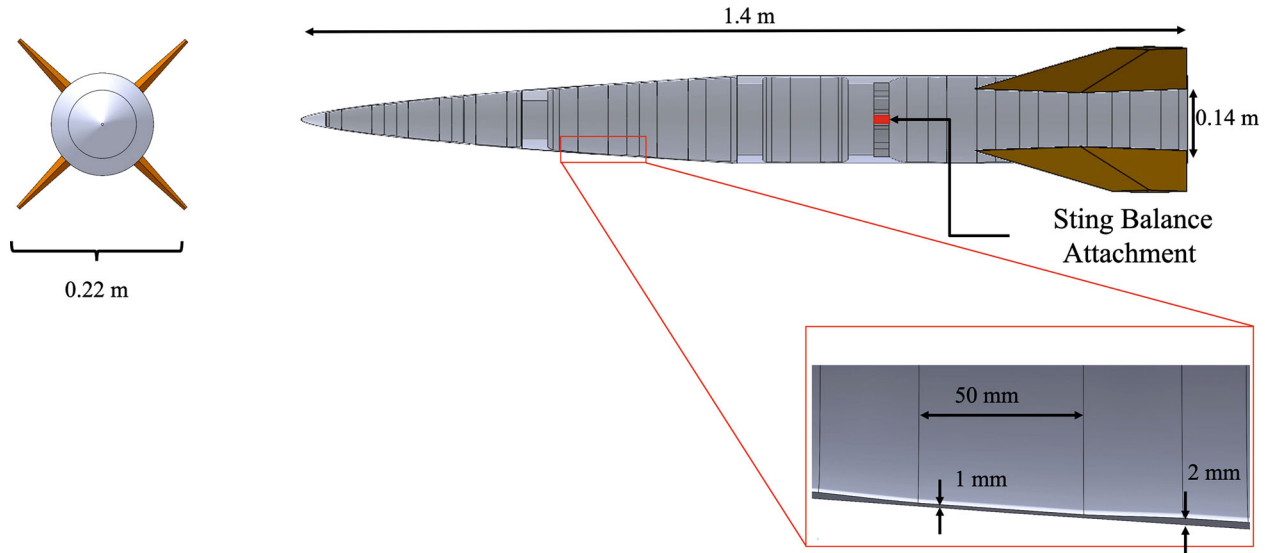


Fig. 2 Cross section of an IC3x nose section, to highlight flexures

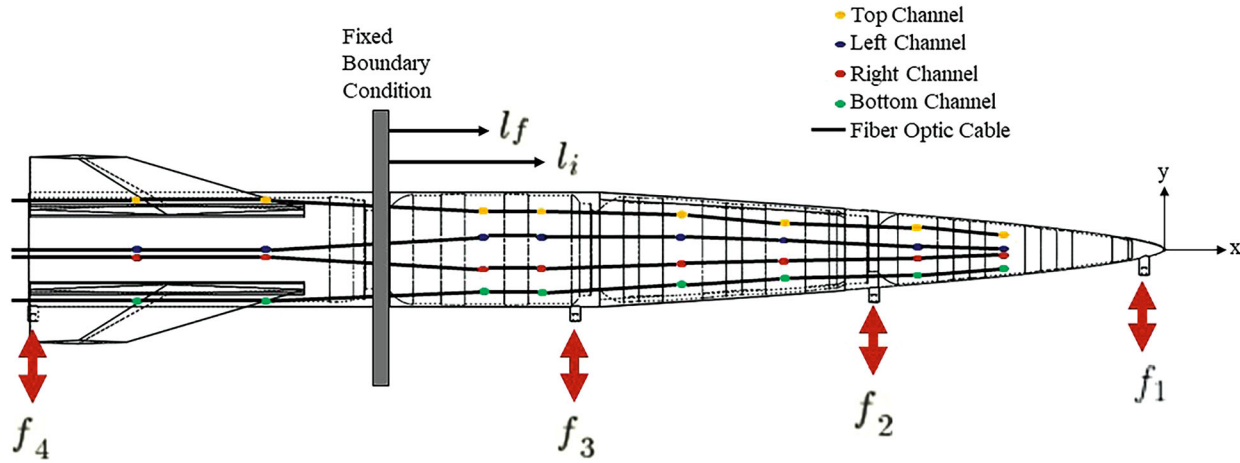


Fig. 3 Graphic representation of sensor longitudinal location, with arrows representing load positions

IC3x Benchtop Model, or IC3x BM, was fabricated with flexures along points of interest where sensors could be fitted with a fiber Bragg grating system. These flexures can be seen in Fig. 2, showing a cross section of the model and highlighting the varied thickness at the flexure. Implementation of these flexures allows a reduction in flexural rigidity, therefore increasing the strain at those locations and improving the signal-to-noise ratio. The material used to fabricate the benchtop model is aluminum ($E = 70 \text{ GPa}$).

Figures 3 and 4, as well as Tables 1 and 2, detail the locations at which the sensors are attached, as well as the channels in which each sensor lies. The method that strain is measured in this experimental setup is a switched Gator fiber optical system that uses Wavelength Division Multiplexing method to measure strain. The channels refer to the sensor lying on one of the four optical fibers utilized in this method, with each optical fiber having eight sensors. To measure strain, point loads were applied to the model at the actuator connective points, as seen in Fig. 3, with specific longitudinal locations described in Table 3. The f in l_f refers to each point load and, therefore, goes from 1 to 4.

Prediction Model

The prediction model implemented to predict bending loads is called the Moore–Penrose inverse matrix, commonly known as the pseudo-inverse matrix model. Given an output vector and input vector, a prediction model based on the pseudo-inverse

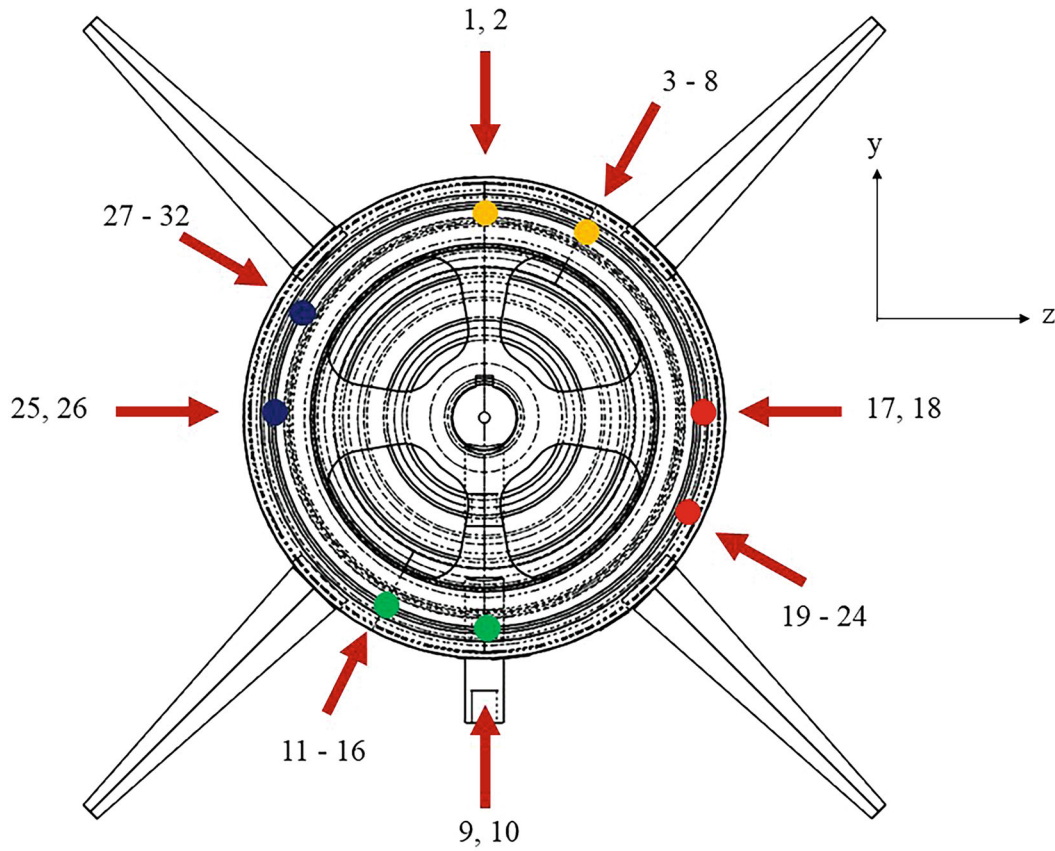


Fig. 4 Graphic representation of sensor angular locations

Table 1 Nominal sensor longitudinal distance from the fixed boundary condition—positive toward the nose—may slightly deviate from experimental setup

Sensor no	l_i (mm)
1, 9, 17, 25	−299.6022
2, 10, 18, 26	−154.8811
3, 11, 19, 27	129.96
4, 12, 20, 28	212.36
5, 13, 21, 29	380.169
6, 14, 22, 30	509.7552
7, 15, 23, 31	676.0128
8, 16, 24, 32	786.0395

Table 2 Nominal sensor angular displacement from the neutral axis in the direction of the positive Y-axis, with positive angular displacements clockwise, as well as sensor channel

Sensor no	θ_{sensor} (deg)	Channel
1, 2	0	Top
3–8	30	Top
9, 10	180	Bottom
11–16	210	Bottom
17, 18	90	Right
19–24	120	Right
25, 26	270	Left
27–32	300	Left

matrix finds an optimum solution that minimizes the norm of the error. Equation 1 shows the form of the pseudo-inverse matrix, where the columns of matrix A correspond to the size of the output vector and the rows correspond to the input vector. $[A]^+$ is a shorthand notation for a pseudo-inverse matrix.

Table 3 Nominal sensor longitudinal distance from the fixed boundary condition—positive toward the nose

Load no	$l_f(mm)$
f_1	961.3471
f_2	620.9414
f_3	244.3625
f_4	−438.9258

$$[A]^+ = (A^T A)^{-1} A^T \quad (1)$$

This chapter aims to predict bending load using bending strain as the input. This means that the matrix A in Eq. 1 has to be a parameter that relates strain and applied loads. The Flexibility Influence Coefficients matrix is one such parameter that relates applied load and strain. In Eq. 2, $[C]$ is the Flexibility Influence Coefficients matrix, ϵ is the bending strain vector where i corresponds to the number of inputs, and f is the force vector (output vector) where a corresponds to the number of outputs. Note that the n is the number of training cases. The generation of the training cases will be discussed in section “[Beam Model and the Generation of Training Data](#)”.

$$\{\epsilon_{i \times n}\} = [C_{i \times a}] \{f_{a \times n}\} \quad (2)$$

The information provided in training comprises the applied load and the resulting strain. This information is the foundation for formulating an expression for determining the Flexibility Influence Coefficients matrix denoted as $[C]$. The process is fundamentally rooted in matrix algebra, which facilitates the Flexibility Influence Coefficients matrix $[C]$. The outcome of this computational procedure is briefly presented in Eq. 3.

$$[C_{i \times a}] = \{\epsilon_{i \times n}\} \{f_{a \times n}\}^T \left(f_{a \times n} f_{a \times n}^T \right)^{-1} \quad (3)$$

Following determining the $[C]$ matrix, the subsequent phase entails formulating a predictive model that utilizes strains as input variables to forecast applied loads as output values. This process is achieved by substituting the matrix A in Eq. 1 with the previously obtained $[C]$ matrix. The ensuing prediction model, employed throughout this chapter, is presented in Eq. 4, where f_p is the predicted load and ϵ is the strain. Notably, the dimensions of the $[C]$ matrix invariably conform to the dimensions $i \times a$. In this investigation, the strain data is recorded from 32 strategically positioned sensors along the IC3x, while 4-point loads are applied to the model as elaborated in section “[Initial Concept 3x \(IC3x\)](#)”.

$$\{f_p\} = (C^T C)^{-1} C^T \{\epsilon\} = [C]^+ \{\epsilon\} \quad (4)$$

Beam Model and the Generation of Training Data

In this specific context, opting for a simpler model offers distinct advantages. It efficiently provides comparable results to the more complex model while maintaining a more intuitive and straightforward structure. The critical concept involves harnessing the local flexural stiffness of sensor placement points to construct a beam model that extends the functionality of the IC3x model.

This process begins by first computing the area moment of inertia at specific sensor locations, followed by its multiplication by Young’s modulus, set at 70 GPa for Aluminum. This results in a beam with varying thickness, where each segment corresponds to the flexural stiffness at the respective strain locations. Figure 5 shows the flexural rigidity of the beam model for two channels. Note that since this is a symmetric geometry, the flexural rigidity remains the same for all other sensors.

Another crucial parameter derived from sensor placement is the distance between the neutral axis and the surface of each subsection of the beam, denoted as y_i . Three parameters remain consistent with the IC3x model: the location of the fixed boundary condition, the position of where various loads are applied (l_f for loads 1, 2, 3, and 4), and the axial distance between the fixed boundary condition and the sensor positions (l_i for sensors numbered 1–32).

$$\{\epsilon_{32 \times t}\} = \sum_{n=1}^4 \frac{f_{n \times t} \cdot (l_f - l_i) \cdot y_i}{EI_i} \quad \text{for } l_i < l_f \quad (5)$$

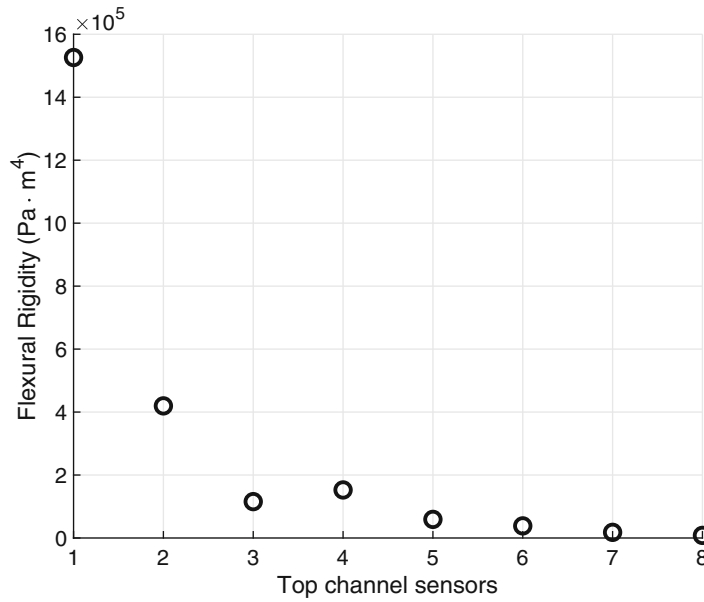


Fig. 5 Flexural rigidity of the beam model

These parameters play a pivotal role in generating training data. A simplified beam model is employed, making it straightforward to calculate strain at each sensor using the Euler–Bernoulli beam equation, as indicated by Eq. 5. This equation relates moments, flexural stiffness, and the vertical distance between the neutral axis and the sensor location to determine the strain. Consequently, the flexibility to generate a substantial amount of training data is achieved by applying a variety of loads in different combinations and computing the corresponding strains. The subscript t denotes the number of the load cases.

In this chapter, each load varies from -150 N to 150 N in increment of 50 N . This signifies that there are seven loads for each loading configuration (f_n), and as there are four-point loads, a total of 7^4 combinations are available for data generation. This data is then used in Eq. 3 to evaluate the $[C]$ matrix, which is then run through Eq. 1 to create a prediction model that predicts point loads when supplied with strain data. The validation procedure of this prediction model is discussed in section “Validation of the Prediction Model” of this chapter.

To assess the accuracy of the beam model in representing the IC3x model, specific load cases were applied to a Finite Element Analysis (FEA) model of the IC3x. The resulting strain from the FEA analysis was compared to the strain computed using Eq. 5. As depicted in Fig. 6, a firm agreement is evident between the calculated strain in the beam model and that in the FEA model. Consequently, it can be confidently asserted that the beam model effectively represents the IC3x model.

Results

Validation of the Prediction Model

Employing equation 5, an extensive dataset of 2401 cases was systematically generated. However, it is worth noting that utilizing the entire dataset is not obligatory. To determine the minimum requisite dataset size, a convergence analysis was conducted focusing on the variable $[C]^+$. This analysis involved systematically varying the size of the training set for predicting point loads, as defined by Eq. 4. Subsequently, the absolute differences between predicted values and actual values for each dataset size are utilized to assess convergence.

The outcomes of this convergence study are visually presented in Fig. 7. This graphical representation unequivocally illustrates that only four training datasets are essential for accurate point load predictions. It should be emphasized that this specific figure corresponds to the convergence study conducted for the f_1 case.

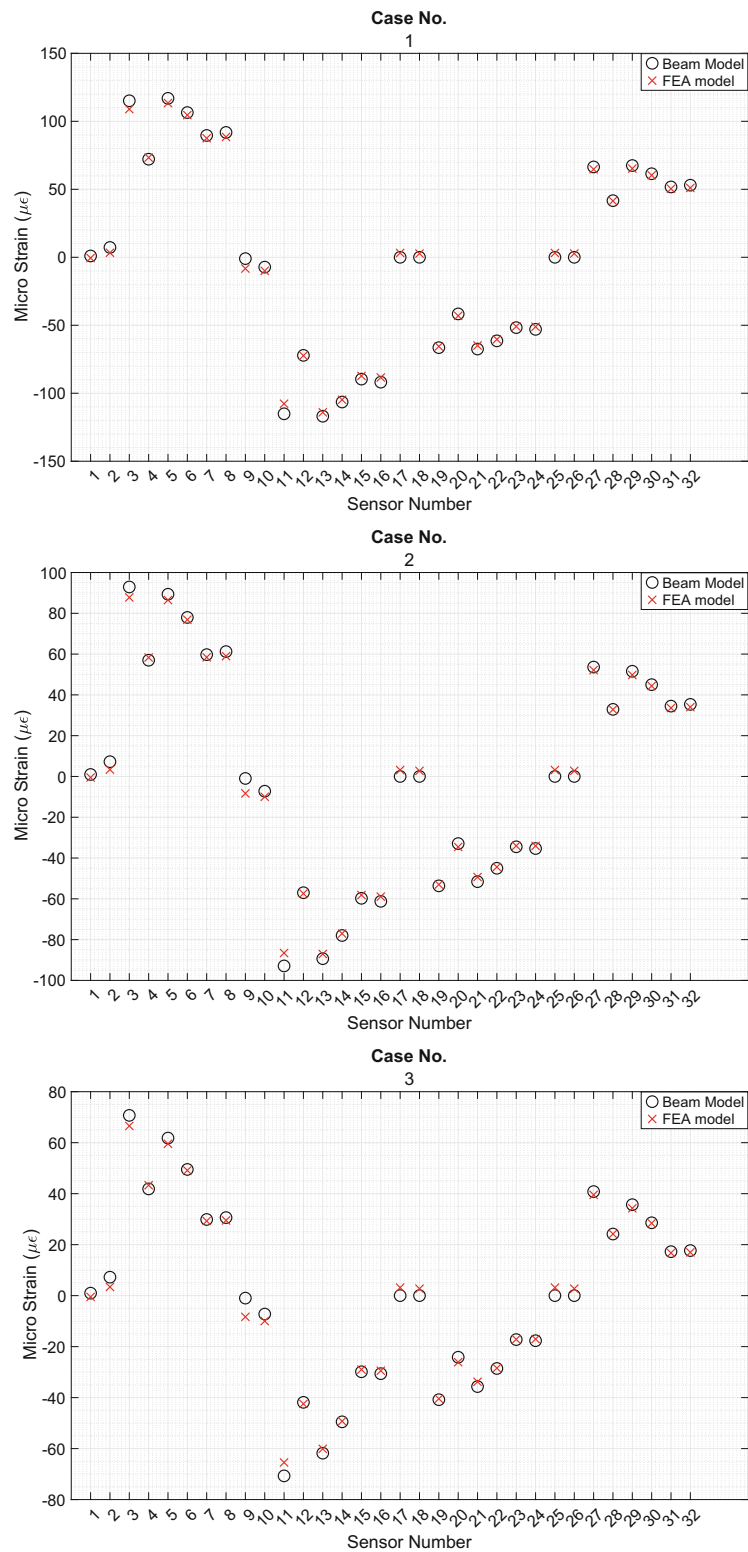


Fig. 6 Comparing strain values between beam model and FEA model

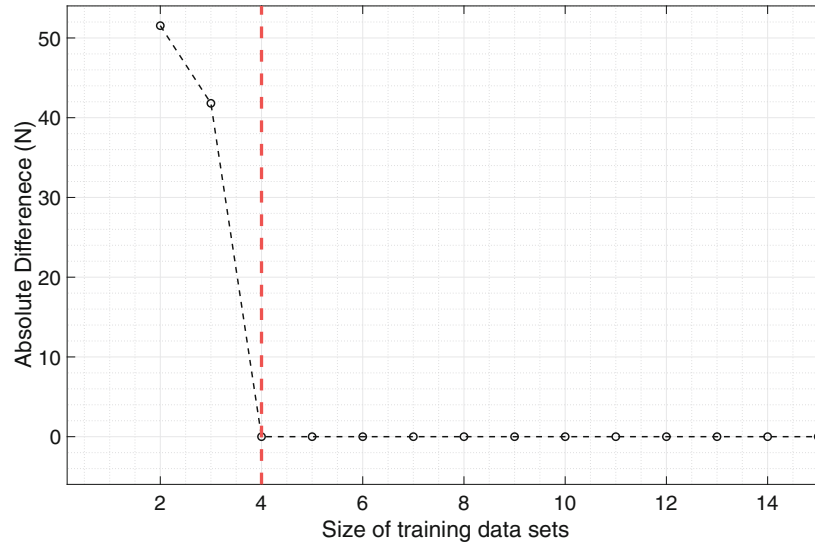


Fig. 7 Convergence study of predicting f_1 with different sizes of training dataset

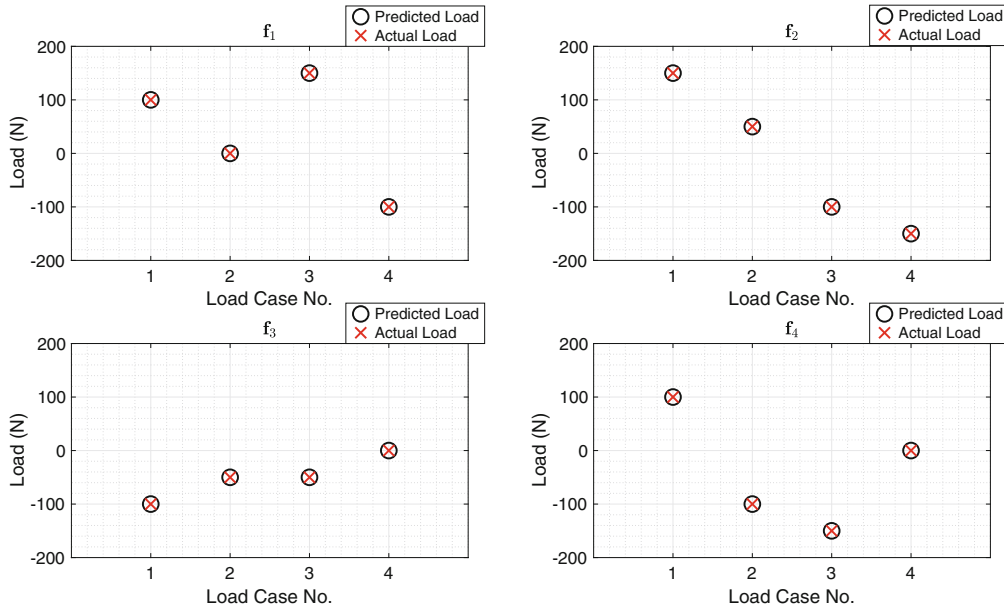


Fig. 8 Validation of the prediction model

The relatively modest number of training datasets required implies that the system is overdetermined, indicating an excess of input variables in comparison to output variables. This observation aligns with the inherent simplicity of the problem, where 32 sensors are employed to predict 4-point loads.

Additionally, Fig. 8 compares the predicted and actual loads for the four selected training datasets. This validation technique is just one of several methods employed to gauge the accuracy of the prediction model when applied to the training data. It is important to note that these four specific training cases will serve as the foundation for the subsequent study in this section.

Testing the Prediction Model with Untrained Test Cases (No Noise)

To assess the prediction model's accuracy, it is imperative to extend our analysis beyond the confines of the training dataset. To accomplish this, a comprehensive set of 221 distinct load cases, each representing a diverse range of combinations,

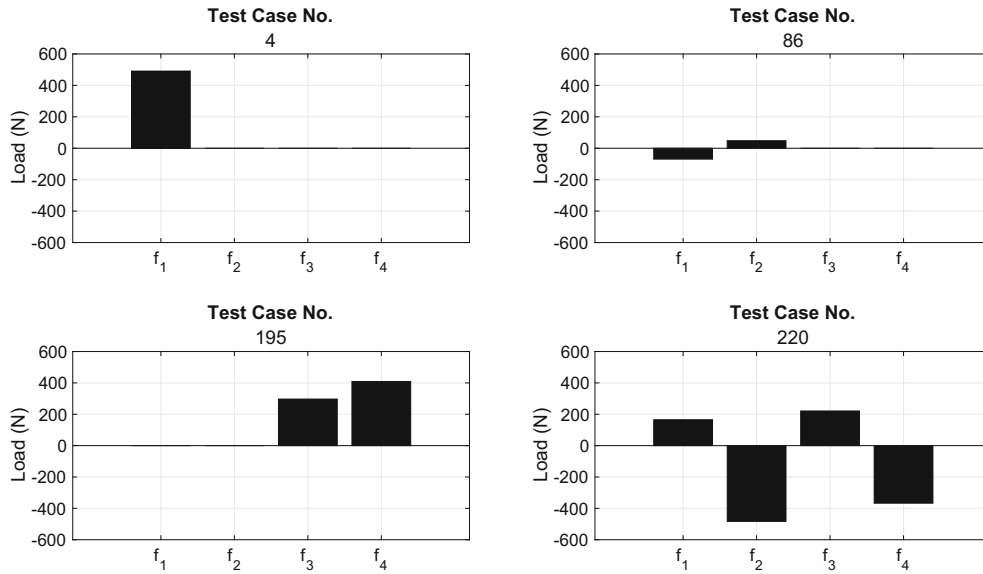


Fig. 9 Sample of point load configuration and value of load from testing set applied to the beam model

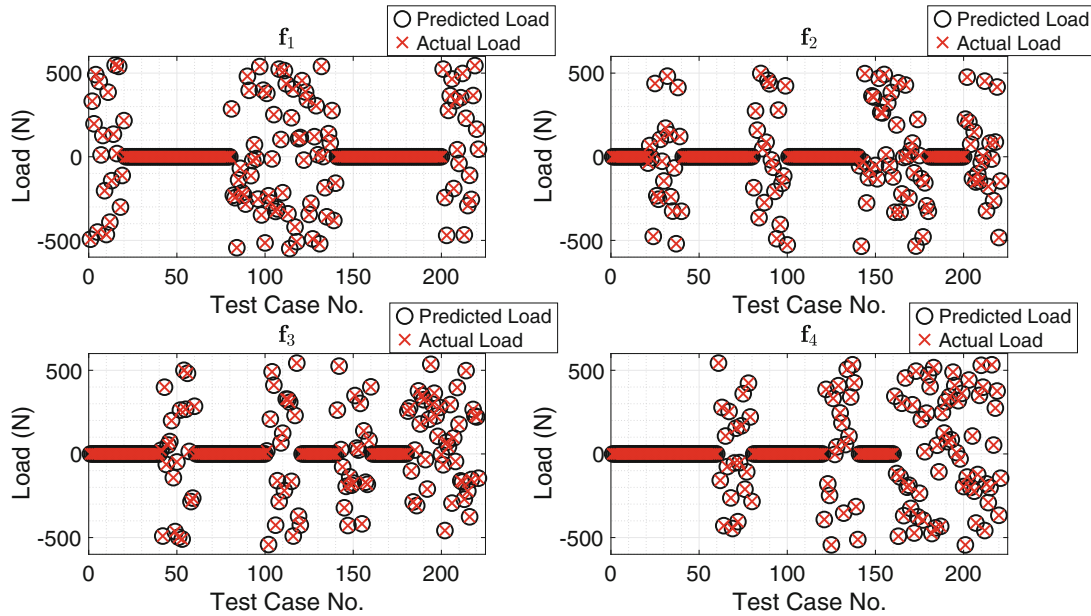


Fig. 10 Comparing predicted load and actual load ($\sigma = 0$)

were generated. Subsequently, we applied Eq. 5 to compute the bending strain corresponding to each of these load cases. This calculated strain served as the input for the prediction model, while the test load cases were systematically examined as variables to gauge the accuracy of the predicted loads. Notably, the testing cases encompassed a broad range of loads, ranging from -500 to 500 N, thereby significantly exceeding the boundaries of the training dataset. There is no synthetic noise added to these test cases. However, a separate study has to be conducted to see the effect of noise on the prediction model's result in section “[Testing the Prediction Model with Untrained Test Cases \(with Noise\)](#)”.

Moreover, Fig. 9 presents four illustrative test cases selected from the extensive set of 221, offering a graphical representation of the diverse load combinations and point load values applied to the beam.

Additionally, in Fig. 10, a comparison is made between the predicted load and the actual load for each f_n . Notably, it is evident from the plot that the predicted load perfectly matches the actual load, demonstrating the accuracy of the prediction model.

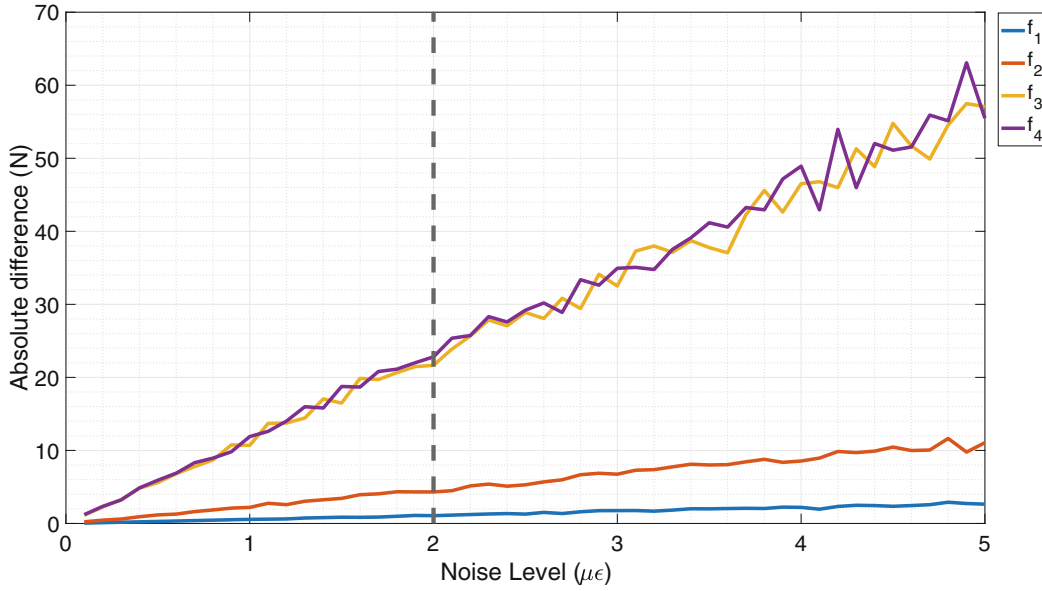


Fig. 11 Effect of different noise levels on the mean absolute difference (δf_n)

Testing the Prediction Model with Untrained Test Cases (with Noise)

The strong agreement observed between predicted loads and actual loads in a noise-free testing scenario presents a promising outcome. However, it is crucial to acknowledge that real-world data inherently contains various degrees of noise. Thus, it becomes imperative to assess the predictive model's performance in the presence of noise, particularly concerning the strain variable.

Since strain measurements are analytically derived using Eq. 5, rather than being experimentally obtained, it is necessary to introduce synthetic noise into the data to simulate real-world conditions. This is achieved through the addition of Gaussian noise to the strain data in the test dataset. The variable ϵ_{test} represents the strain data without any noise. Alternatively, σ quantifies the noise level introduced into the system, enabling the evaluation of the prediction model under various noise levels.

To comprehensively evaluate the model, a range of noise levels from 0.1 microstrain to 5 microstrains is applied to the system, followed by the calculation of the average absolute difference between the predicted load and the actual load for each noise level. This approach provides insights into how the model's accuracy varies with different noise levels. The findings of this analysis are visually represented in Fig. 11, illustrating the impact of noise levels on the prediction model in terms of mean absolute difference. Figure 12 shows the degradation of predicted point loads f_3 and f_4 at 2 microstrains standard deviation.

Specifically, for point loads f_1 and f_2 , there is a gradual increase in the mean absolute difference as the noise level increases. However, the same trend does not hold for point loads f_3 and f_4 . Instead, there is a steep rise in the mean absolute difference, indicating a heightened sensitivity of these point loads to noise. To further evaluate this sensitivity, it becomes imperative to examine the elements within the $[C]^+$ matrix.

Within the $[C]^+$ matrix, a higher value indicates that a specific sensor exhibits greater sensitivity and exerts a more significant influence on predicting the point load. To comprehensively assess the magnitude of each element, it is advisable to refer to the heatmap representation of the $[C]^+$ matrix, as depicted in Fig. 13. This heatmap provides a visual representation of the load's sensitivity to a sensor across different load points. In the heatmap, the rows represent the 4-point loads, and the columns represent the sensor numbers.

It is evident that sensors 3, 4, 5, 11, 12, and 13, corresponding to the top and bottom channels, exhibit higher sensitivity when predicting point load f_3 . Similarly, sensors 1 and 2 (top channel) and sensors 9 and 10 (bottom channel) have a more pronounced influence on predicting point load f_4 .

To explore the factors influencing sensitivity, an analysis of Eq. 5 is required. Within this equation, two constants remain steadfast: the flexural rigidity and the y_i variables. Consequently, attention turns to the remaining variables, namely l_f and l_i . Modifying the value of l_i would alter the foundational beam model itself, thus necessitating its constancy. Consequently,

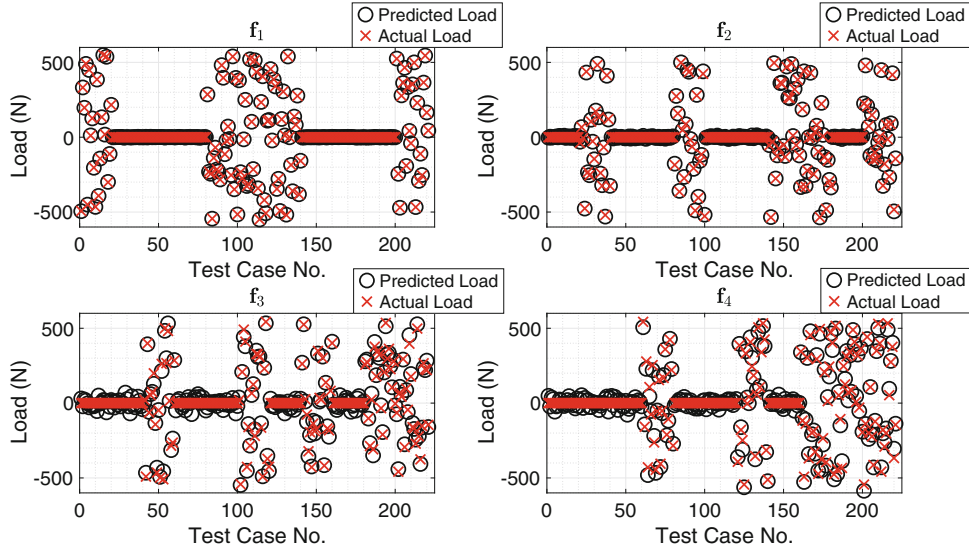


Fig. 12 Comparing predicted load and actual load ($\sigma = 2$ microstrains)

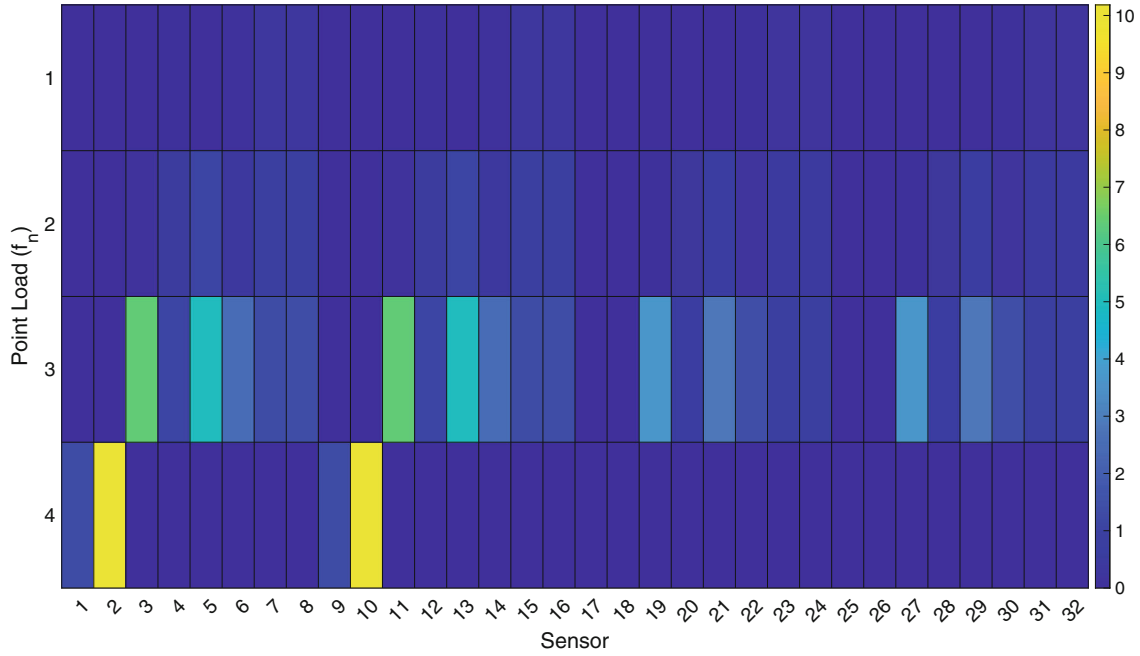


Fig. 13 Heatmap of $[C]^+$

the primary objective of this section centers on the variable l_f , aiming to investigate how fluctuations in its value can impact sensitivity and point load predictions.

To initiate this study, the focus is on altering only l_f of f_3 , extending the distance by 10 cm and decreasing it by 10 cm. Figure 14 illustrates the mean absolute difference of all point loads and heatmap of $[C]^+$ when l_f of f_3 is increased by 10 cm. Figure 15 shows the effect of decreasing l_f of f_3 by 10 cm on the $[C]^+$ matrix and the mean absolute error.

As depicted in Figs. 14 and 15, the manipulation of l_f for f_3 exerts a discernible impact on sensitivity, consequently influencing the susceptibility of load predictions to noise. When the l_f of f_3 approaches the fixed boundary condition, sensitivity escalates, impairing the accuracy of f_3 predictions. Conversely, as the l_f of f_3 moves away from the fixed boundary condition, sensitivity diminishes, leading to a closer alignment between the predicted and actual load values.

During the data collection process in a real experiment, multiple data points are collected within a specific time interval. In the context of static point loads, the data acquisition system records data when the sensors begin to stabilize at a strain value.

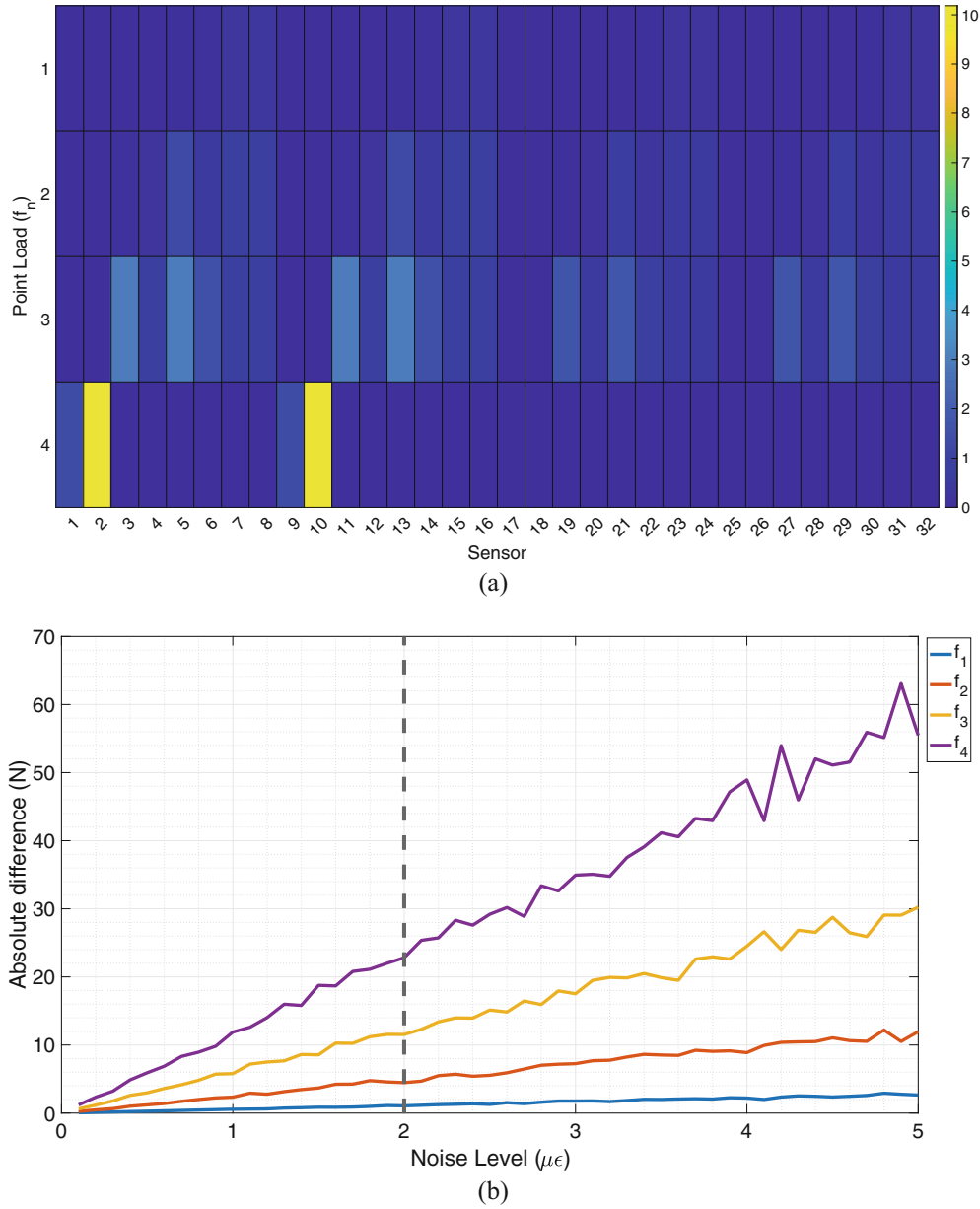


Fig. 14 Effect of increasing l_f of f_3 by 10 cm. (a) Heatmap of the $[C]^+$. (b) Effect of different noise levels on the mean absolute difference

This results in the accumulation of N data points with noise, which can be effectively averaged. This averaging process is instrumental in mitigating the influence of noise, as the average of the noise component tends to converge toward a value of zero. The next portion of the study is to evaluate the effect of N data points on the prediction of the point loads.

Hence, it is essential to examine the prediction model's performance when the number of data points undergoes variation. In this chapter, σ was maintained at a constant value of 2 microstrains. This value was selected due to its minimal impact on predicting f_1 and f_2 , but it had a significant influence on predicting f_3 and f_4 . As Fig. 16 depicts, increasing the number of data points correlates with reducing the mean absolute difference. This finding aligns with the earlier proposition that averaging over a sizable noisy dataset effectively attenuates noise, resulting in a predicted value that closely approximates the actual value.

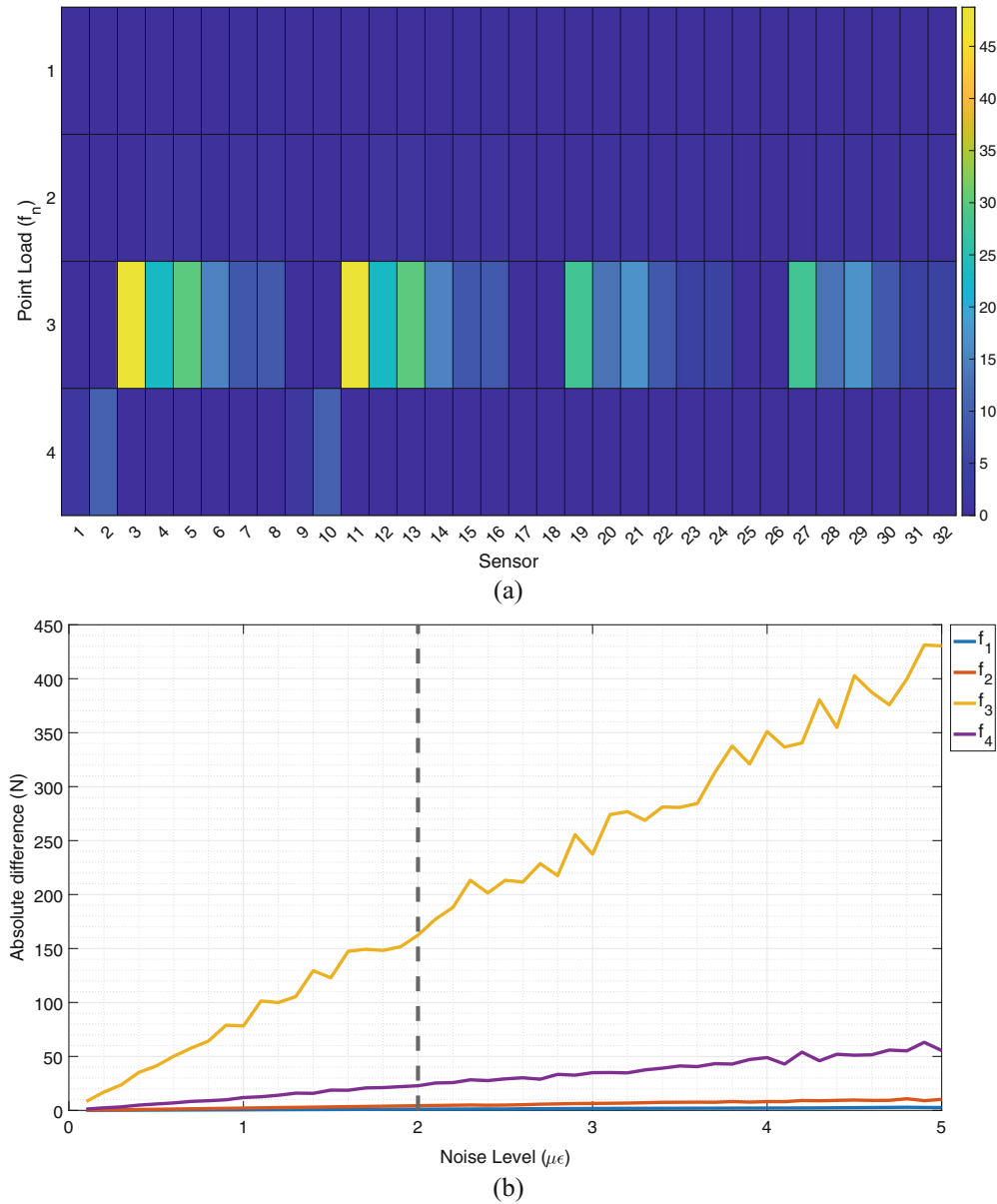


Fig. 15 Effect of decreasing l_f of f_3 by 10 cm. (a) Heatmap of the $[C]^+$. (b) Effect of different noise levels on the mean absolute difference

Effect of Faulty Sensors on Load Predictions

The heatmap of the $[C]^+$ matrix offers an opportunity to investigate the significant role played by sensor locations with heightened sensitivity in predicting point loads. In this section, we examine two real-life scenarios: first, the hypothetical scenario where a sensor with substantial influence on load prediction becomes inoperative, and second, the situation in which a specific sensor malfunctions, resulting in a poor signal-to-noise ratio.

The initial segment of this chapter investigates the impact of sensor removal on point load prediction. Referring to Fig. 14, it is evident that high sensitivity is observed for sensors 1–8 in the top channel and 9–16 in the bottom channel. Our focus in this chapter is solely on the removal of sensors in the top channel. The summary of results can be found in Table 4, where each row corresponds to a removed sensor and its associated mean absolute difference in point load prediction.

As anticipated, the absence of sensor 1 and sensor 2 leads to a deterioration in load prediction for f_4 . For sensors 3–8, any impairment of these sensors negatively impacts the prediction of the f_3 point load. This significant change in results can be attributed to the geometry's inherent symmetry. For instance, when a point load is applied to the model in the positive

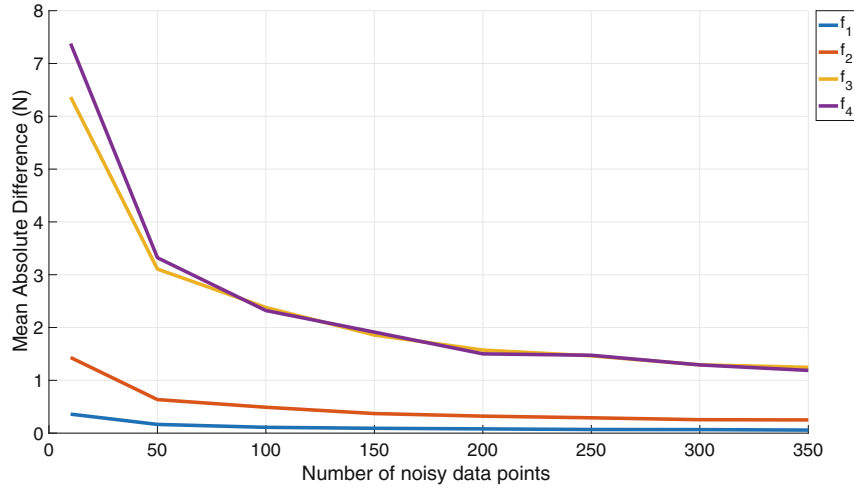


Fig. 16 Effect of increasing noisy data points on the absolute mean difference (δf_n)

Table 4 Effect of sensor removal on the mean absolute difference of f_n [N]

Sensor no	δf_1	δf_2	δf_3	δf_4
1	0.00	0.00	0.00	1.17
2	0.00	0.00	0.00	64.45
3	0.72	10.14	520.85	0.00
4	2.23	31.25	61.52	0.00
5	3.14	101.02	445.24	0.00
6	7.65	32.96	214.28	0.00
7	22.79	62.26	108.46	0.00
8	23.84	65.40	113.94	0.00

y-direction, it induces compression in the top channel and tension in the bottom channel, with equal magnitudes but opposite signs. The deactivation of a sensor in the top channel causes its counterpart sensor in the bottom channel to dominate, resulting in an overestimation of the load, as indicated in Table 4.

The second part of this section addresses the scenario in which a sensor experiences malfunction, providing data with a poor signal-to-noise ratio. To simulate this situation, a 10-microstrain standard deviation Gaussian noise was introduced to each sensor, one at a time, in the top channel. Table 5 presents the findings of this study. As per the findings, the introduction of noise to sensor 2 adversely impacts the predictive accuracy of f_4 . Notably, sensor 4 exerts minimal influence on the degradation of predictions. This phenomenon can be attributed to its distinctive placement within a non-flexure region, leading to reduced sensitivity and, consequently, lower strain values. This observation underscores the role of flexure in enhancing sensitivity by diminishing flexural rigidity. The result of this study is summarized in Table 5.

An intriguing insight derived from this analysis is that the addition of noise to sensors preceding f_n significantly impacts load predictions. For instance, sensors 6 and 5 remain unaffected by f_3 due to their higher l_i values compared to l_f for f_3 . However, the results indicate their substantial contribution to the reduced accuracy of f_3 . An explanation for this phenomenon is provided using the equation that illustrates the prediction model for f_3 when only f_3 is applied to the beam model. All of the sensors would read a 0 strain measurement, except for sensors 3, 4, 11, 12, 19, 20, 27, and 28, as their l_i values are lower than the l_f of f_3 . Ideally, if noise were not added to sensor 6, the equation would only include strains from the sensors mentioned earlier. However, since a high level of noise is added to sensor 6, another term is introduced into the equation, resulting in an overprediction of f_3 . It is worth noting that in β_i^j , i refers to the sensor number, and j represents the point load of interest. In this case, j is equal to 3.

$$f_3 = \beta_4^3 \epsilon_4 + \beta_5^3 \epsilon_5 + \beta_{19}^3 \epsilon_{19} + \beta_{20}^3 \epsilon_{20} + \beta_{27}^3 \epsilon_{27} + \beta_{28}^3 \epsilon_{28} + \underbrace{\beta_6^3 \epsilon_6}_{\text{Faulty Sensor}} \quad (6)$$

Table 5 Effect of adding noise to sensor on the mean absolute difference of f_n [N]

Sensor no	δf_1	δf_2	δf_3	δf_4
1	0.00	0.00	0.00	5.71
2	0.00	0.00	0.00	96.63
3	0.02	0.23	11.56	0.00
4	0.21	2.97	5.84	0.00
5	0.09	3.04	13.41	0.00
6	1.96	8.45	54.95	0.00
7	8.71	23.81	41.48	0.00
8	1.60	4.38	7.63	0.00

Prediction Uniform Temperature Change

The predictive model's versatility goes beyond point load prediction; it can also anticipate temperature fluctuations. When a structure experiences temperature variations, it undergoes either expansion or contraction, inducing strain, even in the absence of external loads. This phenomenon is referred to as thermal strain. The equation below illustrates how to calculate thermal strain based on temperature change, ΔT , and the coefficient of thermal expansion, denoted as α . Since the IC3x model is made out of Aluminum whose α is 24/deg C, the same is applied on the beam model. Since the temperature change is assumed to be uniform throughout the structure, the thermal strain can be added to the strain due to mechanical load.

$$\epsilon_{\text{thermal}} = \alpha \Delta T \quad (7)$$

The process of constructing a prediction model remains consistent with the procedure outlined in section “[Prediction Model](#)”. However, a notable change involves expanding the output vector dimensions from $4 \times n$ to $5 \times n$.

Concerning the training data, the essential requirement for generating C^+ has evolved from a set of 4 to a set of 5, aligning with the principle that the number of training sets should correspond to the dimensions of the output vector. To generate the training data, we compute the thermal strain for a range of temperature changes (-5°C to $+5^\circ\text{C}$), followed by the calculation of the new C matrix. This updated C matrix is then used as depicted in Eq. 4.

For testing the prediction model with the test dataset, the same approach described in section “[Beam Model and the Generation of Training Data](#)” is followed. However, in this updated Eq. 8, there is an additional term representing the thermal strain, as shown in Eq. 8. The term ΔT_n corresponds to 221 testing cases where the values are randomly selected, ranging from -10°C to $+10^\circ\text{C}$.

$$\epsilon_{\text{total}} = \alpha \Delta T_n + \sum_{n=1}^4 \frac{f_{n \times t} \cdot (l_f - l_i) \cdot y_i}{EI_i} \quad \text{for } l_i < l_f \quad (8)$$

Figure 17 compares the predicted temperature change and the actual temperature change for a dataset where a 2-microstrain noise is added to the strain data. It demonstrates good agreement between the actual and predicted temperature change. It is important to note that this aspect of the study is still in its preliminary stages, and further, more comprehensive research is planned to enhance the clarity of predicting mechanical loads in the presence of thermal strain. The mean absolute difference between the predicted temperature change and actual temperature change is 0.011° .

Conclusion

In this comprehensive study, a predictive model tailored for estimating bending loads within structural systems has been developed and rigorously assessed. The core of this model relies on the utilization of strain measurements as input parameters, which are then processed through the pseudo-inverse matrix technique to yield highly accurate predictions of applied loads. This investigation has revealed essential insights into the model's performance across various scenarios, with a particular focus on its sensitivity to noise and its consequent impact on results.

The creation of the beam model was the foundation for load predictions. This analytical model, based on the Euler-Bernoulli beam theory, was meticulously designed to resemble the structural system under investigation.

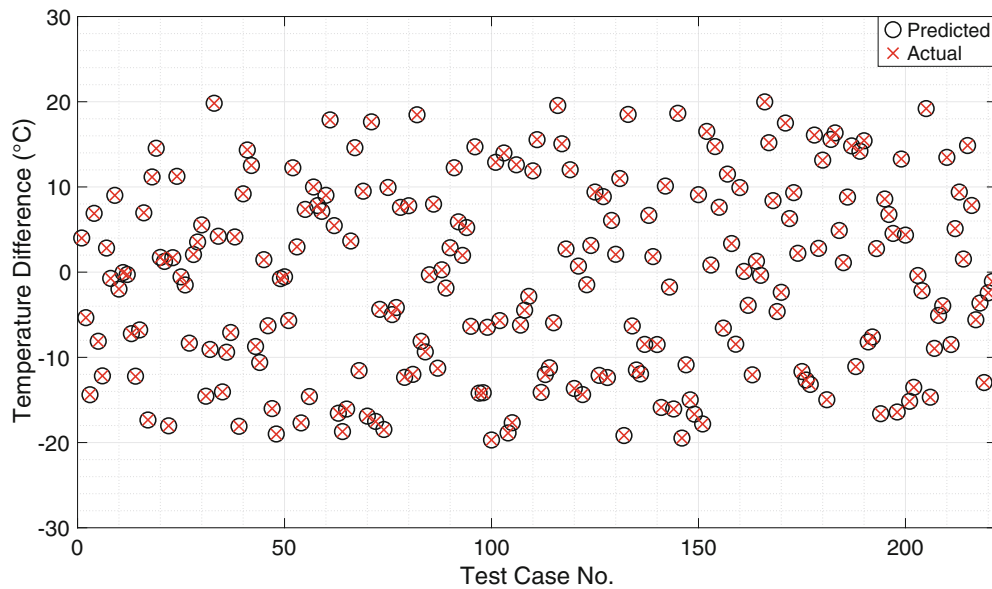


Fig. 17 Comparing predicted temperature difference and actual temperature change

What emerged from this research was a nuanced understanding of sensitivity to noise, which varied across different point loads and sensors. This variability underscored the critical importance of sensor placement and data quality of the given problem. Certain load predictions, notably f_3 and f_4 , displayed heightened vulnerability to noise, as evidenced by a mean absolute difference of up to 60 N at higher noise levels.

These investigations illuminated the model's vulnerability to point load placement (l_f). Shifting l_f of f_3 away from the boundary condition resulted in decreased load sensitivity, leading to a reduced mean absolute difference. Conversely, positioning f_3 closer to the fixed boundary condition heightened load sensitivity, making it more susceptible to noise and resulting in a higher mean absolute difference.

Furthermore, this chapter extensively explored scenarios involving sensor removal and malfunction, exposing their substantial impact on load prediction. When a sensor with significant influence on a specific load condition was removed, load prediction deteriorated. Similar behavior was observed when those particular sensors malfunctioned, leading to an increase in mean absolute difference, indicative of poorer load prediction.

In a notable extension of the model's capabilities, the study delved into its ability to predict temperature changes based on thermal strain inputs. Employing the same procedure for point load prediction, the model effectively predicted temperature changes, achieving close alignment with the actual values.

In summary, the prediction model proves robust in estimating bending loads with close alignment to actual values in a noise-free setting. However, the introduction of noise revealed intriguing insights, highlighting specific loads' sensitivity to particular sensors, which, under noisy conditions, disproportionately influenced load predictions, reducing their accuracy. Moreover, the study emphasized how load sensitivity can be affected by load placement and the detrimental impact of removing or malfunctioning high-impact sensors on predictions. The model's versatility extended to temperature change prediction, showing strong alignment with actual values, exhibiting a minimal mean absolute difference of 0.011 °C.

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